Aeronautics and Space Technology Club

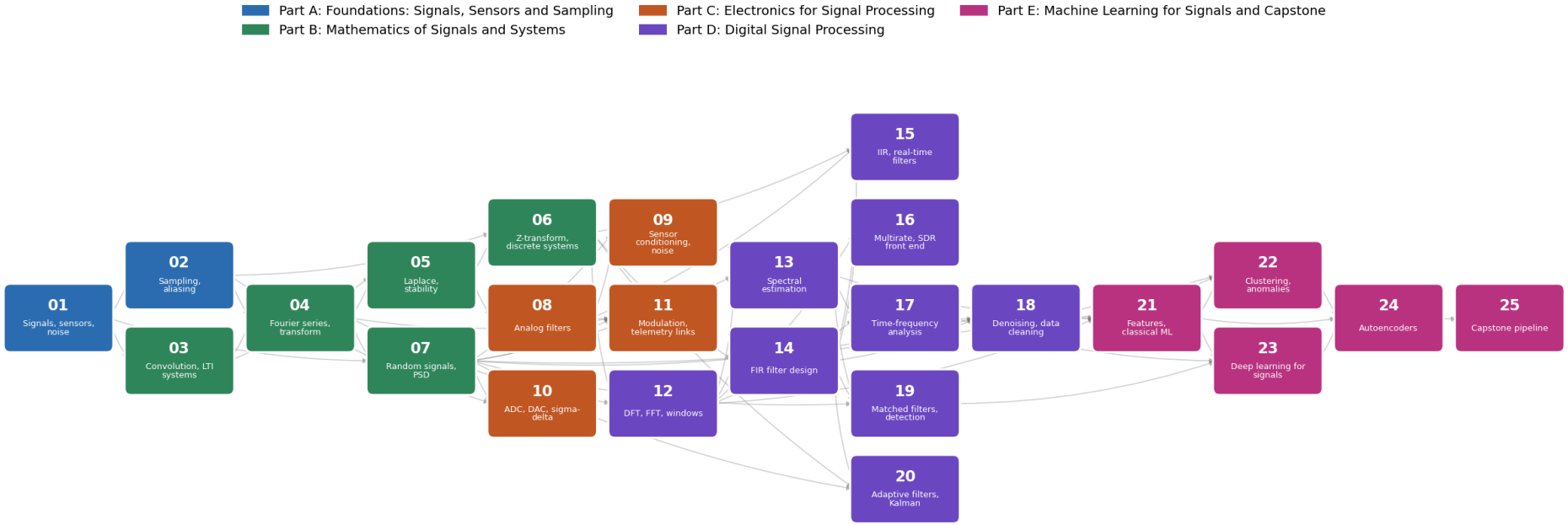
IIT (ISM) Dhanbad

Signal Processing Ladder: Curriculum, Theory and Hands-On Skills

A Ladder of 25 Problem Statements in Signal Processing

Each problem is a self-contained, hands-on problem statement with a synthetic dataset of known truth and pointers to real data. Together they take a student from the first noisy sensor reading to an end-to-end transient-detection pipeline, passing through sampling, the mathematics of signals and systems, analog and digital electronics, DSP, detection and estimation theory and machine learning. Total estimated effort: about 390 hours.

Roadmap



Arrows point from a prerequisite to the problem that builds on it. The capstone (25) needs at least ten earlier problems.

What You Can Do After Each Part

Part A: Foundations: Signals, Sensors and Sampling. You learn to describe a signal by its power and noise, to calibrate a sensor, to separate white noise from drift with the Allan deviation, and to predict, avoid and exploit aliasing. After Part A you can look at any raw sensor stream and say how good it is and at what rate it must be sampled.

Part B: Mathematics of Signals and Systems. You build convolution and LTI-system thinking, the Fourier transform and its uncertainty principle, the Laplace and z-transforms with pole-zero reasoning and stability, and the statistics of random signals including power spectra and Poisson counting. After Part B you can derive, not just use, the formulas of signal processing.

Part C: Electronics for Signal Processing. You design analog filters and verify them with your own circuit solver, size a low-noise sensor front end and a lock-in detector, choose and test ADCs and DACs, and build the physical layer of a satellite telemetry link with its Doppler shift, carrier recovery and link budget.

Part D: Digital Signal Processing. You write the FFT, understand windows and spectral scaling, estimate spectra of noise and of unevenly sampled data, design FIR and IIR filters for real-time use, resample and down-convert radio signals, analyse chirps in time and frequency, clean data without biasing it, build matched filters with calibrated false-alarm rates, and fuse sensors with adaptive filters and Kalman filters.

Part E: Machine Learning for Signals and Capstone. You engineer features and avoid the leakage trap, detect anomalies and bursts with clustering and isolation methods, train convolutional networks and benchmark them against the optimal matched filter, build autoencoders for denoising and novelty detection, and finally assemble an end-to-end transient-detection pipeline with calibrated significance and a report.

Theory and Hands-On Skills by Problem

#	Problem	Theory to master	Hands-on skills gained	Effort
01	**Signals, Sensors and Noise**	Signal classes and measures (energy, power, RMS), decibels, sensor model, noise sources, averaging, Allan variance	NumPy array handling, least-squares calibration, robust statistics, Allan deviation, reading a datasheet noise figure	8–12 hours
02	**Sampling, Aliasing and Reconstruction**	Impulse sampling, spectral replication, Nyquist theorem, aliasing, sinc reconstruction, zero-order hold, bandpass sampling, aperture jitter	Dense-grid simulation of analog signals, alias prediction, interpolation, anti-alias filter selection, jitter Monte Carlo	8–12 hours

#	Problem	Theory to master	Hands-on skills gained	Effort
03	**Complex Numbers, Convolution and LTI Systems**	Complex exponentials and phasors, linearity and time invariance, impulse response, convolution, BIBO stability, correlation, deconvolution	Direct and matrix convolution, numerical LTI tests, system identification from an impulse response, correlation-based ranging, regularised deconvolution	10–14 hours
04	**Fourier Series and the Fourier Transform**	Fourier series, Gibbs phenomenon, Parseval, Fourier transform pairs and properties, convolution theorem, time-bandwidth uncertainty	Numerical Fourier integrals, harmonic analysis and THD, transform-pair verification, uncertainty measurements, spectral reading of rotating machinery	10–14 hours
05	**Laplace Transform, Poles and Stability**	Laplace transform and ROC, transfer functions, poles and zeros, second-order step response, Routh–Hurwitz, Bode plots, PD attitude control	Symbolic transforms with SymPy, pole-zero analysis, step-response metrics, Routh array code, Bode plotting, system identification from a ringing record	10–14 hours
06	**Z-Transform and Discrete-Time Systems**	Z-transform and ROC, difference equations, pole-zero analysis, unit-circle frequency response, discretisation by Euler and bilinear mappings, warping, sample-rate effects on control loops	Difference-equation implementation, pole-zero design of resonators and notches, s-to-z mappings, discrete controller stability analysis, block versus sample processing	10–14 hours
07	**Random Signals, Noise and Power Spectral Density**	Probability distributions, stationarity and ergodicity, autocorrelation, Wiener–Khinchin, filtering of random signals, AR processes, 1/f noise, Poisson counting statistics and detection thresholds	Monte Carlo simulation, correlation estimators, PSD scaling, colored-noise synthesis, Poisson versus Gaussian false-alarm calculation, Fano-factor tests on real counts	12–16 hours
08	**Analog Filters: From RC to Butterworth and Chebyshev**	First- and second-order filters, Bode response, Butterworth Chebyshev Bessel and elliptic families, group delay, Sallen–Key realisation, component tolerance, anti-alias filter specification	Nodal analysis in NumPy, filter order from specifications, pole placement, Monte Carlo tolerance analysis, anti-alias filter design for a given ADC	10–14 hours
09	**Sensor Conditioning and Circuit Noise**	Thermal and shot noise, noise density, op-amp noise, bridge circuits and instrumentation amplifiers, transimpedance amplifiers, lock-in detection, noise budgets	Noise-density to RMS conversion, input-referred noise cascades, bridge and amplifier sizing, digital lock-in implementation, end-to-end noise budgeting	12–16 hours
10	**ADC and DAC: Quantisation and Sigma-Delta**	Uniform quantisation and SQNR, dither, ENOB and SINAD, INL and DNL, oversampling, noise shaping and sigma-delta modulators, zero-order-hold DAC	Quantiser simulation, spectral SINAD and ENOB measurement, code-density test, sigma-delta modulator coding, DAC droop compensation, data-rate budgeting	10–14 hours

#	Problem	Theory to master	Hands-on skills gained	Effort
11	**Modulation, Demodulation and Telemetry Links**	AM and FM, BPSK and matched-filter detection, BER versus Eb/N0, link budget, Doppler shift and rate, carrier recovery with a Costas loop	Modulator and demodulator coding, Monte Carlo BER with confidence intervals, Costas-loop simulation, Doppler curve fitting, link-budget calculation	14–18 hours
12	**DFT, FFT and Windowing**	DFT definition and properties, radix-2 FFT, circular convolution, spectral leakage, window figures of merit, resolution, amplitude and PSD scaling, frequency estimation accuracy	FFT implementation, timing and complexity analysis, leakage and scalloping measurement, correct spectral scaling, peak interpolation for frequency estimation	12–16 hours
13	**Spectral Estimation: Welch, Multitaper and Lomb–Scargle**	Periodogram bias and variance, Bartlett and Welch averaging, confidence intervals, multitaper estimation, coloured-noise slopes, Lomb–Scargle periodogram for uneven sampling, false-alarm probability	Welch implementation and confidence intervals, DPSS multitaper, spectral slope fitting, Lomb–Scargle implementation, period search with aliases and false-alarm probability	12–16 hours
14	**FIR Filter Design**	FIR structure and linear phase, window method, Kaiser design formulas, Parks–McClellan equiripple design, filter types I–IV, differentiators and Hilbert transformers, fast FIR filtering, coefficient quantisation	Windowed-sinc design, order estimation, equiripple design with SciPy, overlap-add filtering, band-pass design on real accelerometer data, fixed-point FIR	12–16 hours
15	**IIR Filters and Real-Time Implementation**	Bilinear transform and prewarping, IIR families, zero-phase filtering, second-order sections and numerical sensitivity, fixed-point arithmetic, causality and detection delay, exponential and complementary filters	Bilinear transform coding, SOS versus direct-form comparison, Q15 biquad implementation, causal event detection with delay and false-alarm statistics	12–16 hours
16	**Multirate Processing and SDR Front Ends**	Decimation and interpolation, polyphase structures, rational resampling, CIC filters and bit growth, digital down-conversion, FM demodulation from IQ, channelisation	Polyphase filter coding, rational resampling, integer CIC implementation, digital down-converter chain, multi-signal channelisation, computational cost accounting	12–16 hours
17	**Time–Frequency Analysis: STFT, Wavelets and Chirps**	Short-time Fourier transform and spectrogram, window-length trade-off, chirps and instantaneous frequency, continuous and discrete wavelet transforms, whitening, gravitational-wave chirp law	STFT and inverse STFT coding, ridge extraction, Morlet CWT via FFT, DWT band analysis, noise whitening, power-law chirp fitting to estimate chirp mass	14–18 hours
18	**Denoising and Data Cleaning for Instrument Data**	Outlier detection, detrending and its bias, Savitzky–Golay smoothing, Wiener filter, wavelet shrinkage, variance-stabilising transforms, gap handling	Robust sigma clipping, Hampel filtering, light-curve flattening with transit masking, Savitzky–Golay coefficient derivation, wavelet thresholding, bias-aware evaluation of cleaning	14–18 hours

#	Problem	Theory to master	Hands-on skills gained	Effort
19	**Matched Filters and Detection Theory**	Neyman–Pearson detection, correlator and matched filter, ROC curves, coloured-noise whitening, optimal SNR, template banks and match, trials factor and false-alarm rate, pulse compression	Monte Carlo ROC estimation, FFT matched filtering, whitening with an estimated PSD, background calibration from noise-only data, template-bank construction, trials-factor calculation	14–20 hours
20	**Adaptive Filters and Kalman State Estimation**	Wiener solution, LMS NLMS and RLS, adaptive noise cancellation, Kalman filter equations, constant-velocity and IMU/GPS models, consistency tests, dead-reckoning drift	LMS and RLS implementation, noise canceller design, Kalman filter coding from scratch, IMU/GPS fusion with bias state, NIS and NEES consistency tests, Allan-variance-based tuning	16–22 hours
21	**Feature Engineering and Classical Machine Learning for Signals**	Segmentation, time-domain spectral and envelope features, bearing fault frequencies, logistic regression SVM and random forests, cross-validation, group leakage, class imbalance, threshold selection	Feature extraction, envelope-spectrum analysis, scikit-learn pipelines, group-aware cross-validation, learning curves, precision-recall analysis and threshold tuning	14–18 hours
22	**Clustering and Anomaly Detection: From n-Sigma to DBSCAN**	n-sigma exceedance detection, DBSCAN and k-distance, Isolation Forest, PCA reconstruction error, injection–recovery efficiency, false-alarm calibration, detector comparison	Event-list binning, baseline and robust sigma, DBSCAN parameter selection, Isolation Forest, injection-recovery experiments, threshold calibration on null data, anomaly detection on spectra	16–22 hours
23	**Deep Learning for Signals: 1D CNNs and Spectrogram Classifiers**	Neural network training, 1D convolution as learned FIR filtering, receptive field, spectrogram CNNs, augmentation, overfitting and learning curves, optimality limits versus matched filtering, calibration and out-of-distribution inputs	PyTorch training loops, 1D and 2D CNN design, receptive-field and parameter counting, augmentation, learning curves, comparison with a matched-filter bank, saliency analysis	16–22 hours
24	**Autoencoders for Denoising and Anomaly Detection**	Linear autoencoders and PCA, convolutional and denoising autoencoders, reconstruction-error anomaly scores, threshold selection, training contamination, bottleneck failure, variational autoencoders	PyTorch autoencoders, PCA equivalence check, windowed time-series anomaly detection, threshold from null reconstruction errors, contamination and capacity studies, comparison with classical detectors	16–22 hours
25	**Capstone: An End-to-End Transient Detection Pipeline**	Pipeline architecture, calibrated detection statistics, trials factors over timescales, injection–recovery completeness, on/off significance, parameter estimation with uncertainties, reproducible reporting	Data audit, cleaning and detection pipeline assembly, two independent detectors, injection-recovery completeness curves, T90 and fluence estimation with bootstrap errors, red-team stress tests, packaging with tests, circular-style report	35–60 hours

Number Hand-Offs Between Problems

From	Used in	What is handed over
01	20, 22, 24	Noise densities and Allan parameters (ARW, bias instability) that set Kalman Q and detector baselines
02	10, 14, 16	Alias prediction, sampling-rate rules, jitter-limited SNR
03	04, 19	Convolution and correlation code, impulse-response identification, delay estimation
04	12, 13, 17	Transform pairs, uncertainty principle, harmonic-line reading of rotating machinery
05	06, 08, 15	Poles and zeros, second-order metrics, the satellite attitude-loop model
06	14, 15, 20	s-to-z mappings, difference-equation implementations, sample-rate limits of discrete loops
07	09, 13, 18, 19, 22	PSD scaling, AR and 1/f noise generators, Poisson statistics and tail probabilities
08	09, 10, 15	Filter families, Sallen-Key design, anti-alias specification
09	10, 20	Noise-budget method, digital lock-in implementation
10	16, 21	ENOB and SQNR methods, the sigma-delta modulator (decimated with a CIC filter in 16)
11	16, 17, 19, 25	Matched-filter BER, Doppler model fit, link-budget calculation
12	13, 14, 17, 21	Window figures of merit, correct amplitude and PSD scaling, peak interpolation
13	17, 18, 19, 21	Welch and multitaper estimators, Lomb-Scargle periodogram, confidence intervals
14	15, 16, 20	FIR design toolbox, overlap-add filtering, delay budgets
15	20, 25	Real-time causal filtering, apogee detector, fixed-point biquads
16	25	Digital down-converter chain, polyphase resampling, CIC filters
17	18, 19, 23	STFT code, whitening pipeline, chirp-law fit
18	21, 22, 24, 25	Bias-aware cleaning pipeline (clip, flatten, denoise)
19	22, 23, 25	Matched-filter SNR series, background calibration, trials factor, ROC tools
20	25	Kalman filter and consistency tests (NIS and NEES)
21	22, 23, 24	Feature library, group cross-validation, threshold selection
22	24, 25	Detector comparison at fixed false-alarm rate, injection-recovery harness
23	24	Training loops and the benchmark against the matched filter
24	25	Autoencoder detector and the contamination study